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Charter

Identify capabilities and areas of potential instability of the "rest api todo list manager".
Identify documented and undocumented "rest api todo list manager" capabilities.
For each capability, create a script or small program to demonstrate the capability.
Exercise each capability identified with data typical to the intended use of the application.
Area: /todos and /todos/:id (45 min)

Session setup

- runTodoManagerRestAPI-1.5.5.jar (v3), started fresh on port 4568
- Windows 10, Java 21, curl in Git Bash, docs at /docs

Summary of session findings

- all documented /todos and /todos/:id operations work (list, filter, HEAD, create JSON/XML, get JSON/XML, amend, delete)
- filters are exact match only
- create validation works (title required, no ids, doneStatus must be boolean) but bools and ids come back as strings
- POST amends, PUT replaces -> wipes fields not sent (docs say amend)
- malformed JSON/XML -> 400 but labelled text/html
- OPTIONS works (not documented), PATCH gives 405
- failed creates still use up ids

Summary of concerns

- PUT loses data if used like the docs say
- error responses have the wrong Content-Type and leak the parser name (gson)
- id gaps, ids never reused -> can't assume ids

Test ideas

- PUT: one test for documented behaviour (fails), one for actual
- malformed JSON/XML tests, check status and Content-Type
- check other todos unchanged after every call
- don't hardcode ids, read them from responses

- next session: relationships, and PUT on a linked todo

Bugs logged

- PUT /todos/:id resets description and doneStatus when not sent (docs: amend)
- Malformed JSON/XML errors are sent with Content-Type text/html

Decisions made during the session

- only /todos and /todos/:id, relationships next session
- fresh app instance, only seed data at start (S1-01)
- only changed todos created in the session, seed todos only read
- set description + done before the PUT to see if it amends or replaces (S1-15 to S1-17)
- ids jumped at S1-10, so checked if failed creates use ids (S1-27/28)
- saved request/response logs instead of screenshots

Areas not covered

- PUT on missing todo, PUT without title
- unknown fields / unknown filters, other Accept types
- PUT/PATCH/DELETE on /todos
- unicode, very long values

Files used or created

- session1_log.txt - all requests and responses (S1-01 to S1-28)
- session1_todos_capabilities.sh + session1_todos_capabilities_output.txt - script for each capability and its output on a fresh app
- runTodoManagerRestAPI-1.5.5.jar

Method	Method Type	is documented ?	Expected Behavior	expected == observed	Observed Behavior	Side Notes	XML & Screenshots
/todos	GET	DOCUMENTED	list all todos	YES	200, 2 seed todos (ids 1, 2)		session1_log.txt S1-01
/todos?doneStatus=false	GET	DOCUMENTED	only matching todos	YES	200, both seed todos		session1_log.txt S1-03
/todos?title=	GET	DOCUMENTED	matching todos	YES	exact title works, partial 'scan' gives []	exact match only	session1_log.txt S1-04/05
/todos	HEAD	DOCUMENTED	headers only	YES	200, no body		session1_log.txt S1-02
/todos	POST	DOCUMENTED	creates todo, assigns id	YES	201, id 3; doneStatus comes back as "false" (string)	bools returned as strings	session1_log.txt S1-06
/todos	POST	DOCUMENTED	no title -> 400	YES	400, title is mandatory		session1_log.txt S1-07
/todos	POST	DOCUMENTED	string doneStatus / id in body -> 400	YES	400 for both	doneStatus has to be a real boolean	session1_log.txt S1-08/09
/todos	POST	DOCUMENTED	create from XML	YES	201, id 6, reply in JSON	id jumped 3 -> 6?	session1_log.txt S1-10
/todos	POST	DOCUMENTED	malformed JSON -> 400	NO	400 but Content-Type text/html, error mentions gson	#BUG	session1_log.txt S1-20
/todos	POST	DOCUMENTED	malformed XML -> 400	NO	400, also text/html	#BUG	session1_log.txt S1-21
/todos	POST	DOCUMENTED	ids in sequence	NO	failed create used up id 7, next todo got 8	#ISSUE	session1_log.txt S1-27/28
/todos	OPTIONS	UNDOCUMENTED	allowed methods	YES	200, Allow: OPTIONS, GET, HEAD, POST	not in docs	session1_log.txt S1-24
/todos/:id	GET	DOCUMENTED	get one todo	YES	200, comes back inside a todos list		session1_log.txt S1-11
/todos/:id	GET	DOCUMENTED	XML with Accept header	YES	200, XML		session1_log.txt S1-12
/todos/:id	GET	DOCUMENTED	missing -> 404	YES	404		session1_log.txt S1-14
/todos/:id	HEAD	DOCUMENTED	headers only	YES	200, no body		session1_log.txt S1-13
/todos/:id	POST	DOCUMENTED	changes only fields sent	YES	200, description changed, rest kept; doneStatus true ok		session1_log.txt S1-15/16
/todos/:id	POST	DOCUMENTED	missing -> 404	YES	404		session1_log.txt S1-19
/todos/:id	PUT	DOCUMENTED	amend (per docs)	NO	200 but description wiped, doneStatus back to false	#BUG acts like replace	session1_log.txt S1-17/18
/todos/:id	DELETE	DOCUMENTED	deletes todo	YES	200		session1_log.txt S1-22
/todos/:id	DELETE	DOCUMENTED	delete again -> 404	YES	404		session1_log.txt S1-23
/todos/:id	OPTIONS	UNDOCUMENTED	allowed methods	YES	200, Allow: OPTIONS, GET, HEAD, POST, PUT, DELETE	not in docs	session1_log.txt S1-25
/todos/:id	PATCH	UNDOCUMENTED	method not allowed	YES	405		session1_log.txt S1-26